Socket Programming Project
Video Streaming using IP Multicast

Description

Implement a multicast video streaming application. The server continuously broadcasts MJPEG video frames to a multicast group, and multiple clients join the group to watch the stream. No RTSP, RTP, TCP/UDP selection, or separate control/data channels are required.

Architecture

Server -> Multicast Group (239.1.1.1:5004) -> Multiple Clients

```
                 +----------------------+
                 |      Video File      |
                 +----------+-----------+
                            |
                            |
                     Read Frame
                            |
                            |
                   +--------v--------+
                   | Multicast Server |
                   +--------+--------+
                            |
          Multicast Group (239.1.1.1:5004)
                            |
        ----------------------------------------
        |                   |                  |
+-------v------+    +------v------+    +------v------+
|   Client A   |    |   Client B  |    |   Client C  |
+--------------+    +-------------+    +-------------+
```

Functional Requirements

Server:

- Read a MJPEG video file frame by frame.
- Packetize each frame.
- Send every frame to a multicast IP address.
- Broadcast frames at approximately 20 FPS (50 ms/frame).
- Continue streaming until the video ends.

- Restart the video automatically after reaching the last frame

Client:

- Join the multicast group.
- Receive multicast packets.
- Decode received packets.
- Display the video in real time.
- Leave the multicast group when exiting.

Running

Server:
python Server.py <file MJPEG>

Client:
python Client.py

Grading Rubric (10 pts)

Requirement	Points	Description
Server implementation	2.5	Multicast server
Client implementation	2.5	Receive/display video
Packet format	2.0	Custom packet
Multiple clients & loss detection	2.0	Concurrency/statistics
Report	1.0	Architecture/testing